Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2023-2024

End-Semester Test
(EC-3 Regular)

Course No.	: DSECLZG529/AIMLCZG529
Course Title	: Data Management for Machine Learning
Nature of Exam	: Open Book
Weightage	: 40%
Duration	: 2 Hours
Date of Exam	: ~~06/03/2021 or 19/03/2021 (FN/AN)~~

No. of Pages = 10
No. of Questions = 7

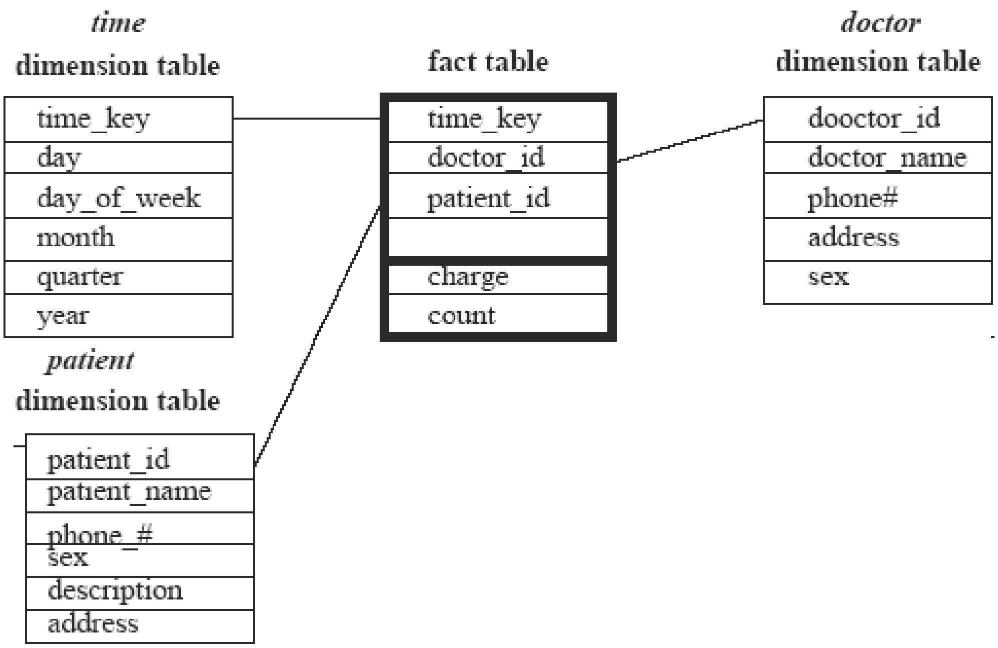
Note to Students:
1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q1. Suppose that a data warehouse consists of three dimensions: **time, doctor,** and **patient**, and the two measures **count** and **charge**, where charge is the fee that a doctor charges a patient for a visit. **[1 + 2 + 2.5 + 1.5 = 7]**

(a) Enumerate two classes of schemas popularly used for modelling data warehouses.
(b) Draw a schema diagram for the above data warehouse using one of the schema classes listed in (a).
(c) Starting with the base cuboid [**day, doctor, patient**], what specific OLAP operations should be performed to list the total fee collected by each doctor in 2004?
(d) To obtain the same list, write an SQL query assuming the data are stored in a relational database with the schema fee (**day, month, year, doctor, hospital, patient, count, charge**).

Answer:

(a) star schema: a fact table in the middle connected to a set of dimension tables
snowflake schema: a refinement of star schema where some dimensional hierarchy is normalised into a set of smaller dimension tables, forming a shape similar to a snowflake.
Fact constellations: multiple fact tables share dimension tables, viewed as a collection of stars, called galaxy schema or fact constellation.
(b) As shown below: Star -schema



Each table 0.5 marks

(c) Steps

1. roll up from day to month to year
2. slice for year = "2004"
3. roll up on patient from individual patient to all
4. slice for patient = "all"
5. get the list of total fee collected by each doctor in 2004

Each step 0.5 marks

(d)

Select doctor, Sum(charge) – 0.5 marks
From fee
Where year = 2004 – 0.5 marks
Group by doctor – 0.5 marks

Q2. You are working on a machine learning project that predicts monthly sales for an e-commerce website. You have trained a sales prediction model using data from the year's first six months (January to June). The model performs well during initial testing.

However, you notice a significant drop in prediction accuracy when you deploy the model in the production environment and start using it to make predictions for the next three months (July to September). You suspect that data drift may be the cause of this decline in performance.

To investigate data drift, you collect the following data for both the training data (January to June) and the production data (July to September):

Training Data (January to June):

- Mean Monthly Sales: $50,000
- Standard Deviation of Monthly Sales: $7,000

Production Data (July to September):

- Mean Monthly Sales: $45,000
- Standard Deviation of Monthly Sales: $8,500

Calculate the following data drift metrics and provide your analysis: **[2 +2 + 2 = 6]**

a) Percentage Change in Mean Monthly Sales between Training and Production Data.
b) Percentage Change in Standard Deviation of Monthly Sales between Training and Production Data.

Answer:

1. **Percentage Change in Mean Monthly Sales:**

   To calculate the percentage change in mean monthly sales, you can use the following formula:

   $$\text{Percentage Change} = \frac{\text{Production Mean} - \text{Training Mean}}{\text{Training Mean}} \times 100$$

   Plugging in the values:

   $$\text{Percentage Change} = \frac{45{,}000 - 50{,}000}{50{,}000} \times 100 = -10\%$$

   The percentage change in mean monthly sales between the training and production data is -10%. This indicates a 10% decrease in mean sales in the production data compared to the training data.

2. **Percentage Change in Standard Deviation of Monthly Sales:**

   To calculate the percentage change in the standard deviation of monthly sales, you can use the following formula:

   $$\text{Percentage Change} = \frac{\text{Production Std Dev} - \text{Training Std Dev}}{\text{Training Std Dev}} \times 100$$

   Plugging in the values:

   $$\text{Percentage Change} = \frac{8{,}500 - 7{,}000}{7{,}000} \times 100 \approx 21.43\%$$

   The percentage change in the standard deviation of monthly sales between the training and production data is approximately 21.43%. This indicates a 21.43% increase in the variability of sales in the production data compared to the training data.

Analysis:

The negative percentage change in the mean monthly sales suggests that the average sales amount in the production data (July to September) is 10% lower than in the training data (January to June). This could be due to changes in customer behavior, economic factors, or other external influences.

The positive percentage change in the standard deviation of monthly sales indicates that the variability in sales has increased by approximately 21.43% in the production data compared to the training data. This suggests that the sales data in the production environment may be more volatile or unpredictable, which can significantly impact the performance of the sales prediction model.

To address the data drift issue, you may need to retrain your model using the latest data or implement strategies to adapt the model to the changing patterns in sales data.

Q3. You are tasked with designing a data pipeline to process and analyse log data from a website. The logs contain user interactions, including page views, clicks, and demographics. The pipeline consists of three stages: data ingestion, transformation, and analysis.

- Data Ingestion: The raw log data is ingested into the pipeline at an average of 1,000 log entries per second.
- Data Transformation: During the transformation stage, various data processing tasks are performed, including data cleaning, parsing, and feature extraction. The transformation stage processes data at 800 log entries per second.
- Data Analysis: In the analysis stage, machine learning models are applied to the transformed data to predict user behavior. The analysis stage processes data at an average of 500 log entries per second.

a) Calculate the data throughput for each stage of the data pipeline in log entries per minute. **[3]**
b) Determine the bottleneck stage in the data pipeline based on the calculated throughputs. **[1]**

Answer:

a) Calculating Data Throughput:

To calculate the data throughput for each stage, we'll first calculate the processing rate per second for each stage and then convert it to log entries per minute.

1. **Data Ingestion:**
   - Processing Rate (per second) = 1,000 log entries/second
   - Processing Rate (per minute) = 1,000 log entries/second * 60 seconds/minute = 60,000 log entries/minute
2. **Data Transformation:**
   - Processing Rate (per second) = 800 log entries/second
   - Processing Rate (per minute) = 800 log entries/second * 60 seconds/minute = 48,000 log entries/minute
3. **Data Analysis:**
   - Processing Rate (per second) = 500 log entries/second
   - Processing Rate (per minute) = 500 log entries/second * 60 seconds/minute = 30,000 log entries/minute

b) Determining the Bottleneck Stage:

The bottleneck stage is the stage with the lowest throughput because it determines the overall capacity of the data pipeline.

In this case, the bottleneck stage is the **Data Analysis** stage, as it has the lowest throughput of 30,000 log entries per minute. This means that even if the other stages can process data at a higher rate, the Data Analysis stage cannot process data faster than 30,000 log entries per minute, limiting the overall pipeline throughput.

To optimize the data pipeline's overall throughput, you may need to consider various strategies, such as parallel processing, distributed computing, or hardware upgrades, to improve the performance of the bottleneck stage or balance the processing rates across all stages.

Q4. Imagine you are a data scientist working on a machine learning project for a healthcare organisation. Your task is to build a predictive model to identify patients at high risk of developing a specific medical condition based on their health records.

In your machine learning project, you trained three models: Model A, Model B, and Model C, each with various hyperparameters and feature engineering techniques. As part of your model metadata, you have recorded the model's architecture, hyperparameters, training data, evaluation metrics, and the date of each model's creation. Additionally, you have documented any noteworthy observations or lessons learned during the modelling process.

You are reviewing the model metadata for Models A, B, and C, and you notice that Model C consistently outperforms the other models in terms of accuracy and recall on the validation dataset. However, Model C is also significantly larger regarding memory usage than Models A and B. Given this information:

a) Why is it important to keep track of model metadata in your machine learning project? **[2]**

b) What are the potential advantages of using Model C despite its higher memory usage? **[2]**

c) How can you ensure that the large memory usage of Model C is smooth in a production environment? **[1]**

Answer:

a) Keeping track of model metadata is crucial for several reasons:

- Reproducibility: Model metadata allows you to recreate and reproduce the same model in the future, which is essential for auditing and validation.
- Performance Evaluation: Metadata helps you compare different model iterations, understand which models perform better, and make informed decisions about model selection.
- Troubleshooting: When issues or unexpected behaviour arise, model metadata provides historical context and insights into what might have caused the problem.
- Documentation: Metadata serves as documentation for your machine learning pipeline, making it easier for collaborators to understand and work with your models.

b) The advantages of using Model C, despite its higher memory usage, may include:

- Higher Accuracy: Model C consistently outperforms the other models regarding accuracy and recall, which may lead to more accurate predictions and better patient risk assessment.
- Improved Patient Care: The improved performance of Model C may help healthcare providers identify patients at higher risk more accurately, allowing for early interventions and better patient care.
- Research Insights: The larger model may capture more complex patterns in the data, potentially leading to valuable insights or discoveries about the medical condition.

c) To ensure that the large memory usage of Model C does not become a bottleneck in a production environment, you can consider the following strategies:

- Model Optimization: Explore techniques for model optimisation, such as model pruning, quantisation, or using more memory-efficient architectures.
- Hardware Scaling: Assess the organisation's hardware capabilities and, if necessary, invest in hardware upgrades or cloud resources that can accommodate larger models.
- Batch Processing: Implement batch processing for predictions if real-time processing is not required, as this can help manage memory usage.
- Monitoring and Scaling: Continuously monitor memory usage in production and set up auto-scaling mechanisms to handle increased loads as needed.
- Model Compression: Explore model compression techniques that reduce memory requirements while retaining most of the model's performance.

By considering these strategies, you can leverage the advantages of Model C while mitigating potential challenges related to its higher memory usage in a production healthcare environment.

Q5. You are tasked with designing a data architecture for a large e-commerce platform. The platform handles many daily customer transactions, product updates, and user interactions. The architecture must efficiently support real-time analytics, reporting, and data storage. You decide to use a data warehouse for this purpose.
Here are some critical metrics for the platform:

- 10,000,000 customer transactions per day
- 1,000,000 product updates per day
- 5,000,000 user interactions per day

Your data architecture needs to handle and process this data efficiently. Design a data architecture that includes the following components and provide an estimate of the required storage capacity: **[4]**

- Data Ingestion Layer
- Data Storage Layer
- Data Processing Layer
- Data Analytics and Reporting Layer

Answer:
To design a data architecture for the e-commerce platform, we need to consider each layer and estimate the required storage capacity.

1. **Data Ingestion Layer:**
   - In this layer, data from various sources (customer transactions, product updates, user interactions) is ingested into the system.
   - Given the metrics:
     - 10,000,000 customer transactions per day
     - 1,000,000 product updates per day
     - 5,000,000 user interactions per day
   - The daily data ingestion rate is 10,000,000 + 1,000,000 + 5,000,000 = 16,000,000 records per day.
2. **Data Storage Layer:**
   - In this layer, data is stored efficiently. We need to estimate the required storage capacity based on daily data ingestion and retention policies.
   - Assume each record on average is 1 KB in size.
   - The daily data ingestion is 16,000,000 records per day.
   - If we want to retain data for a year (365 days), the annual storage requirement would be:
     - 16,000,000 records/day * 1 KB/record * 365 days = 5,840,000 GB or approximately 5.84 petabytes (PB) of storage per year.

3. **Data Processing Layer:**
   - In this layer, data is processed, transformed, and prepared for analytics and reporting.
   - The required processing capacity depends on the complexity of the transformations and the desired processing speed. It may involve distributed processing using technologies like Apache Spark or Hadoop.
4. **Data Analytics and Reporting Layer:**
   - In this layer, data is used for real-time analytics and reporting.
   - The performance and responsiveness of this layer depend on factors like query complexity, indexing, and hardware infrastructure.

In summary, for this data architecture, you would need approximately 5.84 petabytes of storage capacity per year to handle the daily data ingestion and retain data for a year. The other layers would require scalable and efficient technologies to handle data processing, analytics, and reporting volume and complexity.

Q6. Consider a dataset containing information about customer orders for an e-commerce website. Here are 20 records with various data quality issues. Identify at least 5 data quality issues and compute data quality metrics to help illustrate the importance of data quality assessment. **[10]**

```
Order_ID,Order_Date,Product,Quantity,Total_Price,Shipping_Address
101,2022-01-05,Widget,3,$45.00,123 Main St, Cityville
102,2022-01-10,Widget,5,$75.00,,Cityville
103,2022-02-15,Widget,2,$30.00,456 Elm St, Townsville
104,2022-02-20,Widget,1,$15.00,789 Oak St, Villageville
105,,Widget,4,$60.00,234 Birch St, Townsville
106,2022-03-25,Widget,2,$30.00,567 Pine St,
107,2022-03-30,Widget,3,$45.00,890 Cedar St, Cityville
108,2022-04-05,Widget,,,$70.00,123 Main St, Cityville
109,2022-04-10,Widget,5,$, Townsville
110,2022-05-15,Widget,3,$45.00,123 Main St, Cityville
111,2022-05-20,Widget,-2,$30.00,456 Elm St, Townsville
112,2022-06-25,Widget,2,$35.00,789 Oak St, Villageville
113,2022-06-30,Widget,3,$,, Cityville
114,2022-07-05,Widget,4,$65.00,234 Birch St, Townsville
115,2022-07-10,Widget,2,$30.00,567 Pine St, Villageville
116,2022-08-15,Widget,1,$15.00,, Villageville
117,2022-08-20,Widget,4,,$60.00,789 Oak St, Villageville
118,2022-09-25,Widget,3,$45.00,890 Cedar St, Cityville
119,2022-09-30,Widget,,,$70.00,123 Main St, Cityville
120,2022-10-05,Widget,5,$75.00,123 Main St, Cityville
```

Answer:
Data Quality Issues:

- Missing values in "Shipping_Address" field (rows 2, 5, 6, 9, 13, 16, 18).
- Missing values in "Total_Price" field (rows 4, 9, 13, 17).
- Negative quantity in "Quantity" field (row 11).
- Invalid date format in "Order_Date" field (row 5, 13).
- Duplicate Order_IDs (rows 1 and 10 have the same Order_ID).

Data Quality Metrics:

1. **Percentage of Missing Values in Shipping_Address:**
   - Number of missing values in "Shipping_Address" = 7
   - Total number of records = 20
   - Percentage of missing values = (7 / 20) * 100 = 35%
2. **Percentage of Missing Values in Total_Price:**
   - Number of missing values in "Total_Price" = 4
   - Total number of records = 20
   - Percentage of missing values = (4 / 20) * 100 = 20%
3. **Percentage of Negative Quantity Values:**
   - Number of records with negative "Quantity" values = 1
   - Total number of records = 20
   - Percentage of negative values = (1 / 20) * 100 = 5%
4. **Percentage of Records with Invalid Date Format:**
   - Number of records with invalid date format = 2
   - Total number of records = 20
   - Percentage of records with invalid date format = (2 / 20) * 100 = 10%
5. **Percentage of Duplicate Order_IDs:**
   - Number of duplicate Order_IDs = 1 (Order_ID 101 is duplicated)
   - Total number of records = 20
   - Percentage of duplicate Order_IDs = (1 / 20) * 100 = 5%

Q7. A healthcare organisation is sharing medical research data with a research partner while ensuring privacy protection using a privacy-preserving technique called "k-anonymity." The dataset contains information about patients' medical conditions and their ages. The organisation wants to disclose aggregate statistics about patient ages while protecting individual privacy.
The organisation chooses to achieve 3-anonymity, meaning that there are at least three patients with the same age and medical condition for any combination of age and medical condition. The dataset contains the following information:

- Patient A: Age 45, Medical Condition X
- Patient B: Age 30, Medical Condition Y

- Patient C: Age 45, Medical Condition Z
- Patient D: Age 35, Medical Condition X
- Patient E: Age 50, Medical Condition Y
- Patient F: Age 45, Medical Condition X
- Patient G: Age 30, Medical Condition Z
- Patient H: Age 30, Medical Condition X

a) Calculate the transformed dataset that satisfies the 3-anonymity requirement. **[3]**
b) Explain how the transformed dataset ensures privacy protection for individual patients. **[1]**

Answer:
a) To achieve 3-anonymity, we need to transform the dataset so that for any combination of age and medical condition, there are at least three patients with the same age and medical condition. We can do this by generalising and grouping patients with similar attributes.
The transformed dataset for 3-anonymity:

- Group 1: Ages 30-35, Medical Condition X
  Patients: B, D, H
- Group 2: Age 45, Medical Condition X
  Patients: A, F, (Any other patient with age 45 and Medical Condition X)
- Group 3: Age 30, Medical Condition Y
  Patients: E (Any other patient with age 30 and Medical Condition Y)
- Group 4: Age 45, Medical Condition Z
  Patients: C (Any other patient with age 45 and Medical Condition Z)
- Group 5: Age 30, Medical Condition Z
  Patients: G (Any other patient with age 30 and Medical Condition Z)

We've grouped patients with similar ages and medical conditions in the transformed dataset. Each group satisfies the 3-anonymity requirement because at least three patients are in each group.
b) The transformed dataset ensures privacy protection for individual patients by obscuring their specific attributes. Here's how it provides privacy:

- Individual Patient Identification: Identifying a specific patient within a group is difficult because multiple patients share the same age and medical condition. For example, in Group 1, patients B, D, and H share similar attributes, making it challenging to distinguish one from another.
- Preventing Attribute Disclosure: The transformed dataset prevents attribute disclosure by only revealing generalised information, such as age ranges and medical conditions, for groups of patients. This reduces the risk of exposing sensitive patient information.
- Meeting the Anonymity Requirement: By ensuring 3-anonymity, the organisation guarantees at least three patients with the same age and medical condition in each group. This makes it statistically challenging to re-identify individual patients from the disclosed data.

Overall, k-anonymity techniques like this one balance the need for sharing aggregated data for research purposes while protecting the privacy of individual patients.

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Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2023-2024

Mid-Semester Test
(EC-2 Regular – ANSWER KEY)

Course No.	: DSECLZG529/AIMLCZG529
Course Title	: Data Management for Machine Learning
Nature of Exam	: Open Book
Weightage	: 30%
Duration	: 2 Hours
Date of Exam	: ~~06/03/2021 or 19/03/2021 (FN/AN)~~

No. of Pages = 5
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3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q1. Imagine that you are building a next-generation telecom company, one that help users to avoid wastage of data from their existing data package subscription. First step is to deploy smart agent at customers' phone. These agents will report the application data use of the phone every hour, along with information on which applications are consuming the more data. Based on this, we would like to offer customers variable pricing in real time, based on when and how they run their favorite applications. **[1 + 2 + 1 + 1 + 2 = 7]**

a) Show a sample of the tuple generated out of the agent on phone.
b) Give two examples of exploratory analysis you will do on this data? Give details.
c) What can be the target variable in this example?
d) What sort of ML modelling technique can be utilized for it?
e) Which type of system architecture will be useful in this scenario? Draw a suitable architecture diagram for the same.

Answer:
a) {Timestamp, phone_no, {app1:data_usage, app2:data_usage, app3:data_usage,…}}
b) Each applications hourly / daily / day wise (week-day/week-end) data usage
   Day wise data requirements for a device
   Categorizing the data usages by types of applications – entertainment, work related
c) Recommended plan or subscription value
d) Classification – for plan recommendation , Regression – for plan value
e) Lambda – Both current and historical data taken into consideration for plan recommendation

Q2. Consider a data storage for University students. Each student data, stuData, which is in a file of size less than 64 MB (1 MB = 2^{20}B). A data block (of default size 64MB) stores the full file data for a student of stuData_idN, where N =1 to 500. Assume that the cluster used to store these records consists of racks which has two DataNodes for processing each of 64 GB (1 GB = 2^{30}B) memory. Cluster consists of 120 racks. Replication factor 3. **[1 + 3 + 2 + 1 + 1 = 8]**

a) How the files of each student will be distributed at a cluster?
b) How many students' data can be stored at one cluster?
c) What is the total memory capacity of the cluster (in TB) and DataNodes in each rack?
d) How many racks will be required if only 5000 student's data needs to be stored?
e) What will be the changes when a stuData file size <= 128 MB?

Answers:

a) How the files of each student will be distributed at a cluster?
   Answer: Data block default size is 64 MB. Each students file size is less than 64 MB. Therefore for each student file one data block will suffice.

b) How many students' data can be stored at one cluster?
   Answer:
   A data block is in a DataNode. Each rack has two DataNodes each of memory size 64 GB.
   Each node can store 64 GB / 64 MB = 1024 data blocks = 1024 student files
   Each rack thus store 2 * 1024 data blocks = 2048 student files

   Each block default replicates three times in DataNodes.
   Number of students whose data can be stored in the cluster
   = number of racks * number of files / 3 = 120 * 2028 / 3 = 81920

c) What is the total memory capacity of the cluster (in TB)?
   Answer:
   Total memory capacity of cluster
   = 120 * 128 GB = 15360 GB
   = 15 TB

d) How many racks will be required if only 5000 student's data needs to be stored?
   Answer:
   One rack will store 2048 students' data
   For storing 5000 students data = 5000 / 2048 = 2.44 racks = 3 racks will be required (without replication)
   With replication = 3 * 3 = 9 racks will be required

e) What will be the changes when a stuData file size <= 128 MB?
   Answer: Changes will be that each node will have half number of data blocks.

Q3. An organization currently has an on-premise installation of an overall database system. This organization keenly looks into the developments in database systems, especially cloud databases. **[2 + 3 + 2 + 1 = 8]**

a) Name the type of cloud service that will match the requirement of hosting a database in the cloud. Enlist two cloud services from prominent vendors that can be helpful here.
b) Enlist three significant advantages the organization will achieve by migrating to the cloud.

c) List two changes that users of this system will face by migrating the databases to the cloud.
d) If the organization looking to support the analytical queries, what will be your suggestion for them?

Answer:

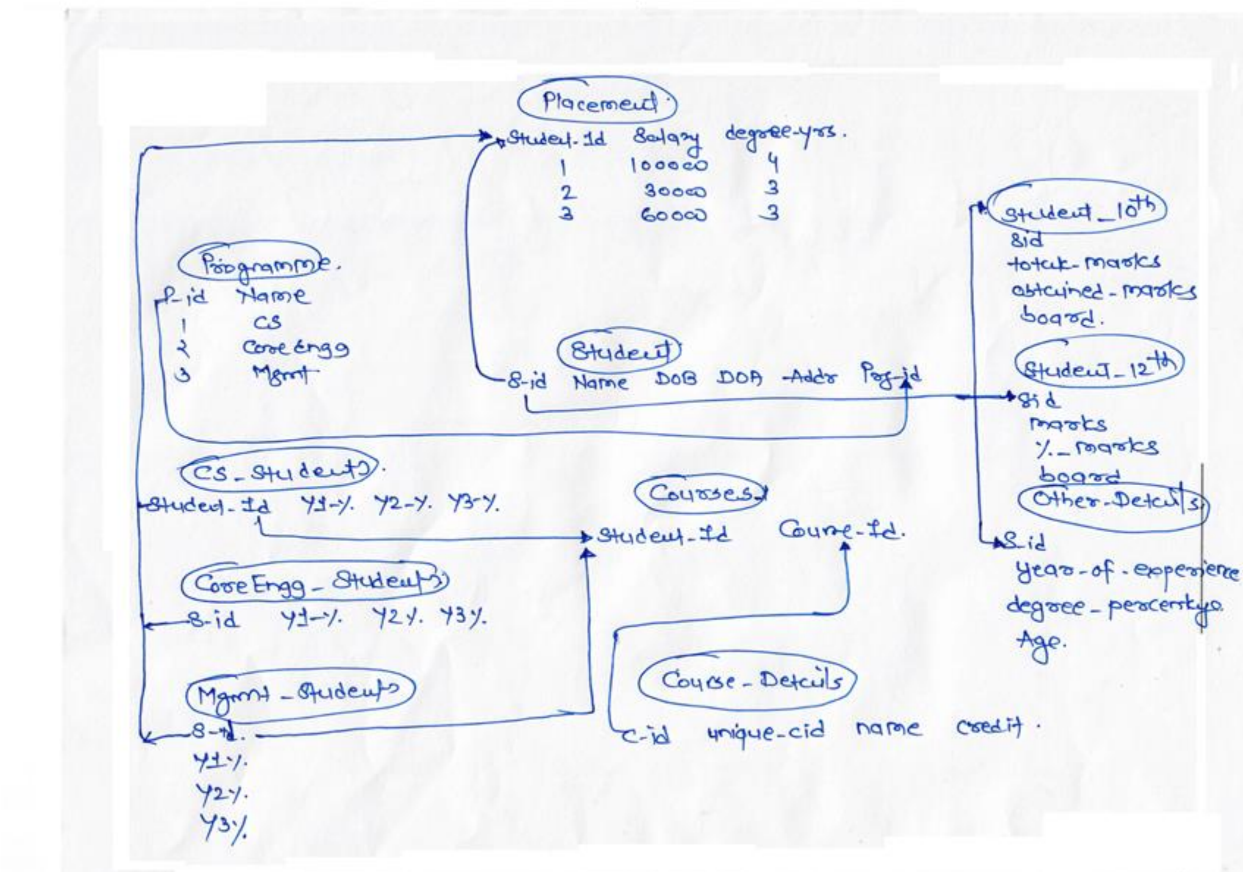
a) Database as a service - 1M
AWS RDS, Azure SQL database – 1M
b) Three advantages by migrating to the cloud – Identifying each advantage 1 mark
   i. Elimination of physical infrastructure- In a cloud database environment, the cloud computing provider of servers, storage and other infrastructure is responsible for maintenance and keeping high availability. The organization that owns and operates the database is only responsible for supporting and maintaining the database software and its contents. In a DBaaS environment, the service provider is responsible for managing and operating the database software, leaving the DBaaS users responsible only for their own data.
   ii. Cost savings- Through the elimination of a physical infrastructure owned and operated by an IT department, significant savings can be achieved from reduced capital expenditures, less staff, decreased electrical and HVAC operating costs and a smaller amount of needed physical space.
   iii. DBaaS benefits also include instantaneous scalability, performance guarantees, failover support, declining pricing and specialized expertise.

c) Two changes that users of this system will face by migrating the databases to the cloud – Identifying each change 1 mark
   i. Always internet connection will be required to access the database
   ii. Response time for queries might increase

d) Use any of cloud warehouse for data storage and analytics queries such as AWS Redshift, Google BigQuery, Snowflake etc.

Q4. The dean of placement division of Premiere Institute has collected data on their recent placement. To attract good students, it is important for the school to ensure that the students are placed with good salary package. The dean believed that the salary earned by a student at placement dependent on several variables.
The data collected by Dean has the following structure associated with it (look at next page). But he is not able to identify the significant attributes that are helping the students to earn good salaries. You are consulted by him to help in this exercise.

a) List down two major ambiguities which are present in the given ER diagram.
b) Flatten out the data set by applying the general data preprocessing techniques. Show a sample record of the same
c) Using the flattened out dataset, derive a model (just based on the intuition) that will predict what range of salaries can be earned by the student? **[2 + 4 + 1 = 7]**



Answer:

Answer:

a) Which student id needs to be considered? It's referred differently all over the places. What is the duration of the programmers? The year wise marks, degree marks are scattered.

Each ambiguity -1 mark, 1 * 2 = 2 marks

b) Flattened data set

Student_id	Name	Age	DOA	Addr	Programme	SSC marks	SSC Board	HSC marks	HSC board
Year_Of_Exp	Degree_perc	Y1 %	Y2 %	Y3 %	Y4%	Degree_years	Salary		

Removal of duplicate attributes – 1 mark
Removal of irrelevant attributes – 1 mark
Converting the attributes into appropriate forms – 1 mark
Sample record – 1 mark

c) Student can derive logistic regression model just taking two salary ranges into consideration like HIGH or LOW

Or they can think of building a decision tree where the outcome can be discrete salary ranges like LOW, AVERAGE, and HIGH etc.

Any one of the above model – 1 mark

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No. of Pages = 6
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Q1. Assume that you are working on a Real Estate app that allows the users to Search New Projects, Homes, Apartments, Offices, Shops, and Showrooms for Buy, Sell, and Rent. This property app is required for fulfilling all the real estate needs for searching, shortlisting and finalizing property of their choice. Be it a ready-to-move flat in a new project, investment in shops, offices or showrooms, or girls PG or boys PG in city, app should have listings from qualified property agents & top builders as well as no brokerage properties. This house search app should make property search on mobile simpler, faster and smoother. Search for residential and commercial property listings from owners, brokers and top builders on this user-friendly app should help millions of users to start their real estate journey. To make it more interesting and meaningful, when the users are online searching for a property on the app, the recommendations about the relevant properties should be shown to her. Also relevant advertisements has to be shown on the app screen to increase the chances of converting these paid advertisements into increasing the foot falling and result into actual selling. **[3 + 2 + 1 + 1.5 + 2 + 2.5 + 1 + 2 = 15]**

a) Who are the primary users of this systems and what type of user interfaces are suitable for them to interact with this application?

Answer:

- Property Owners – Web / Mobile interfaces
- Property Searchers – Web / Mobile interfaces
- Real-Estate Agents – Web / Mobile interfaces / APIs
- Portal Executives – Web / Mobile interfaces / APIs

b) Discuss the important data modalities of data that are suitable for such applications?

Answer:

- Structured data – user profiles
- Semi-structured data – property details

Assume a builder who is promoting his real-estate project on this application needs a periodic report on the searches happened for the project, type of project information accessed, brochure downloads, queries raised against property etc.

c) What type of data abstraction will be required to generate such a report? Why?

Answer:

- Data Warehouse / Data Mart – 0.5M
- as most of the data is structured data which can be stored in RDBMS backed DW / DM – 0.5M

d) What type of data processing is suitable for the above requirement? Why?

Answer:

- Batch processing - 0.5M
- as it needs to be generated periodically – 1M

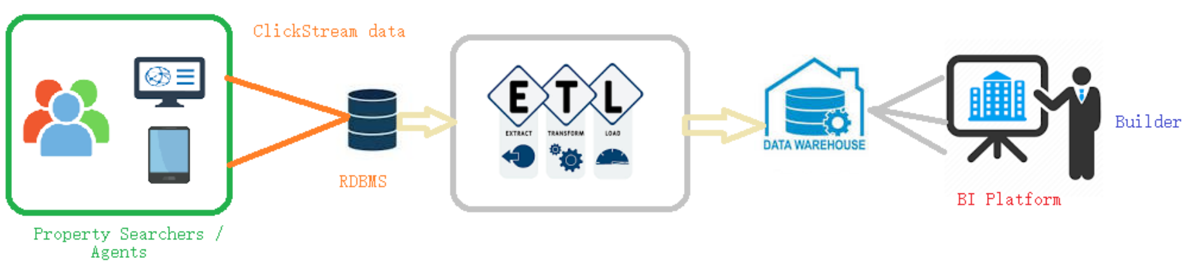
e) Identify any two approaches through which this prepared report can be served to the builder or his representative?

Answer:

- Through web browser / BI application interface
- Report that can be downloaded in CSV, Spreadsheet or pdf format
- Report link that can be shared over email or other communication channels

f) Discuss briefly the data flow (and the stages involved) that will be required to meet this requirement along with a pictorial representation.

Answer:



Components

- Users
- OLTP System – RDBMS
- ETL pipeline
- Data Warehouse
- BI Platform

Each component – 0.5M

Imagine that now you have to implement a feature where immediate property recommendations needs to be shown to the users when they are looking for a property.

g) What is the significant change will be required in dataflow mentioned in (f) to match the new requirement?

Answer:

- Streaming storage and streaming processors needs to be implemented in the dataflow.

h) Which data format will be suitable to exchange the data between users of the application and recommendation engine? Provide a snippet of the data item exchanged.

Answer:

- JSON will be a suitable data format for data exchange. 1M
- Input – JSON structure having user applied filters and some user profile data - 0.5M
- Output – JSON list of properties matching to the property user is looking upon - 0.5M

Q2. Predictive Automotive Components Services (PACS) Company renders customer services for maintenance and servicing of (Internet) connected cars and its components. Assume that number of centers are 8192 ($=2^{13}$), number of car serviced by each center per day equals 32 ($=2^5$). Each car has 256 ($=2^8$) components, which requires maintenance or servicing in the Company's car. The service center also collects feedback after every service and send responses to customer requests. The feedback and responses text takes on average 128 B ($=2^7$ B) and each service or responses records in a report of average 512 B ($= 2^9$) text. Company saves the centers data for maximum 10 years and follows last-in first-out data replacement policy. **[1 + 1.5 + 1.5 + 1 + 2 = 7]**

a) Justify why PACS is example of big data use case.
b) What are the big data system characteristics that you think will be important in this service?
c) How will the files of PACS be saved using big data file system like GFS or HDFS?
d) If the data block size is 64MB, how many records will be stored per block?
e) What shall be the minimum memory requirement (in MB) for 10 years?

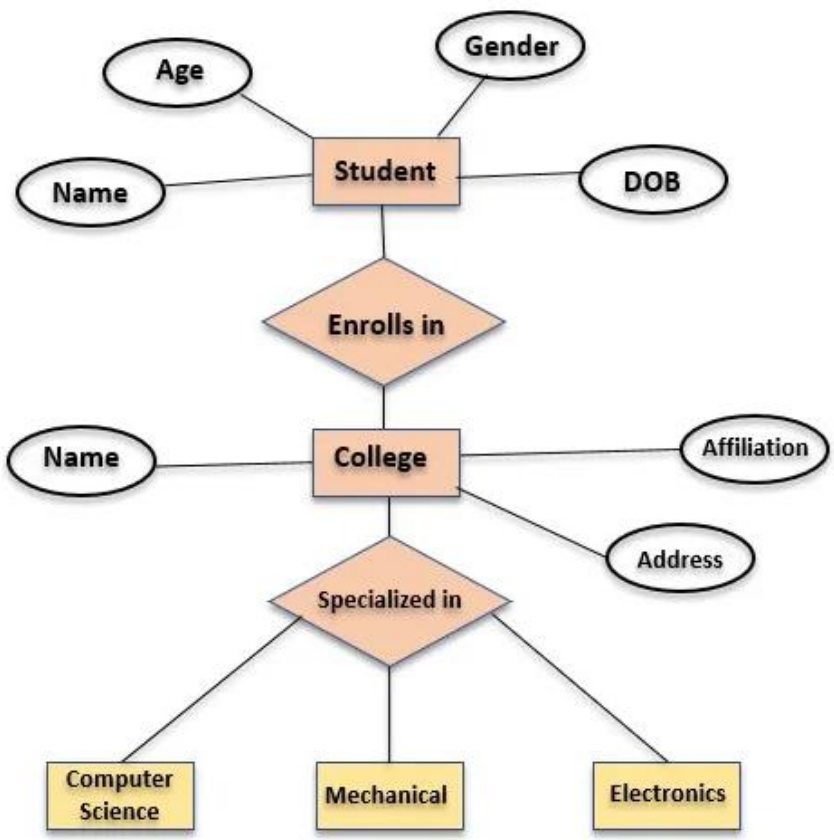
Answer:

a) Its big data use case as it caters to big data's "Volume" property.
b) Big data system characteristics: scalability, reliability and maintainability
c) The HDFS will use the following mechanism to store the file
    - NamedNode will identify the DataNodes on which data needs to be stored
    - Data is stored in data blocks on the DataNodes
    - HDFS replicates those data blocks and distributes them so they are replicated within multiple nodes across the cluster.
d) Data files size per car = 512B (Service Report) and 128B (feedback) = 640B (512B+128B)
   Therefore, each data block may store 64MB/640B that is ~ 100K records.
e) The minimum memory requirement = [No. of centers * No. of cars serviced per day per center * No. of days in a year * 10 years * (Size of feedback + size of Service Report)] B

$= [2^{13} * 2^{5} * 365 * 10 * 640]$ B
= 584000 MB

Q3. Consider the following Entity-Relationship Diagram representing the relationship between students and college. Represent this ERD in the following data format using sample data. **[3 + 2 + 1 + 2 = 8]**

I. Row-major format
II. Columnar format
III. CSV
IV. JSON



Answer:

I. Row-major format – RDBMS
RDBMS relation – three tables – student, college and Specialization

Student

StudentID	Name	Age	DOB	CollegeID
1	A	21	1-jan-2001	1

2	B	23	1-march-2001	2

College

CollegeID	Name	Affiliation	Address	SpecializationID
1	SBC	Pune	Pune	1
2	BITS	Pialni	Pilani	2

Specialization

SpecializationID	Name
1	CS
2	Mech

II. Columnar format – NoSQL stores like Cassandra – will use flat structure – store all column values next two each other

Student Name	Age	DOB	CollegeName	Affiliation	Address	Specialization Name
A	21	1-jan-2001	SBC	Pune	Pune	CS
B	23	1-march-2001	BITS	Pialni	Pilani	Mech

```
{
        StudentName: {A, B},
         Age:{ 21, 23},
        DOB:{ 1-jan-2001, 1-march-2001 },
        COllegeName:{ SBC , BITS },
        Affiliation:{ Pune , Pialni },
        Address:{ Pune , Pilani },
        Specialization Name{ CS , Mech }

}
```

III. CSV - – can embed all the data in a single file

```
StudentName, Age, DOB, College Name, Affiliation, Address, Specialization
A,      21,     1-jan-2001,     SBC,    Pune,   Pune,   CS
B,      23,     1-march-2001,BITS,   Pialni, Pilani, Mech
```

IV. JSON

```
{
        {
                Name: A,
                Age: 21,
                DOB: 1-Jan-2001,
                College :{
                        Name: S B C,
                        Affiliation: Pune ,
                        Address: Pune ,
                        Specialization: CS
                }
          },
        {
                Name: B,
                Age: 23,
                DOB: 1-march-2001,
                College :{
                        Name: S B College,
                        Affiliation: BITS,
                        Address: Pilani,
                        Specialization: Mech
                }
          }

   }
```

****